Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 2^{nd} Year 1^{st} Term Examination-2023
Math 2119
Mathematics III

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a) Define the following matrices with examples: **(i)** Lower triangular matrix, **(ii)** Equivalence matrix, **(iii)** Hermitian matrix, **(iv)** Trace of a matrix. 10

1(b) Express the matrix $A=\begin{bmatrix} 1 & 2 & 9 \\ -2 & 5 & 7 \\ 15 & 102 & -31 \end{bmatrix}$ as a sum of symmetric and skew-symmetric matrices. 08

1(c) Define inverse of a matrix. Find the inverse matrix of $A=\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, if it exists. 11

1(d) Define idempotent matrix. Find the value of x, y, z and a which satisfy the matrix equation $\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4a-6 \end{bmatrix}=\begin{bmatrix} 0 & -7 \\ 3 & 2a \end{bmatrix}$. 06

2(a) Define rank and normal form of a matrix. 06

2(b) Reduce the following matrix to echelon form, then to canonical form and then to normal: 17
$A=\begin{bmatrix} 2 & 4 & 3 & 2 \\ 3 & 6 & 5 & 1 \\ 2 & 5 & 2 & -3 \\ 4 & 5 & 14 & 14 \end{bmatrix}$. Also, find its rank.

2(c) Solve the following system of equations $\begin{aligned} x_1-2x_2-x_3+3x_4&=1 \\ 2x_1-4x_2+x_3&=5 \\ x_1-2x_2+2x_3-3x_4&=4 \end{aligned}$. Write your solution in parametric form. 12

3(a) Define Laplace transform. Find $L\{t^4-5e^{5t}+9\cos 2t-3e^{2t}\sin 3t\}$ 10

3(b) Let F(t)=3t for 0<t<2 and F(t)=6 for 2<t<4, where F(t) has period 4, find L{F(t)}. 12

3(c) Solve $y'-4y=1;\ y(0)=1$ using Laplace transform. 13

4(a) Define inverse Laplace transform. Compute $L^{-1}\left\{\frac{4e^{-\pi s}}{s^2+4s+20}\right\}$. 10

4(b) Find $L^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}\right\}$. 10

4(c) Solve $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$; U (x, 0) = $3\sin 2\pi x$, U (0, t)=0, U (1, 0)=0 where $0<x<1$, $t>0$. 15

SECTION–B

5(a) Define periodic function and its fundamental period. Find the fundamental period of $f(x) = \sin^2 \pi x$. Aiso, sketch the graph of f(x). 10

5(b) Define the inner product between functions. Find the inner product of cosx and $\sin^2 x$ and determine whether they are orthogonal or not over $[0, \pi]$. Also find the norm of $\sin^2 x$ over the given interval. 12

5(c) Sketch the periodic function $g(x) = x+1,\ 0 < x < 1$ defined here on a single period. Hence find the Fourier series of $g(x)$. Give the number to which Fourier series of g(x) converse at $x = 1/2$. 13

6(a) Find the Fourier half-range cosine series of the function $F(t) = 2t,\ 0 < t < 1$
$= 2(2-t),\ 1 < t < 2.$ 13

6(b) Sketch odd and even extension of $i(x) = e^{-x}; 0 < x < 1$. Hence, find its Fourier cosine series. 12

6(c) Find the complex Fourier series of $p(x) = \begin{cases} 5\ ;\ 0 < x < \pi \\ -5;\ \pi < x < 2\pi \end{cases}$ where p(x) has the property of $p(x) = p(x+2\pi)$. 10

7(a) Give three physical examples of multivariable scalar functions. Sketch the vector field $\vec{V}(x^2 - y^2)$. 10

7(b) Find the directional derivative of the divergence of $f(x,y,z) = xy\,\hat{i} + xy^2\hat{j} + z^2\hat{k}$ at the point (2, 1, 2) in the direction of the outer normal to the sphere $x^2+y^2+z^2=9$. 12

7(c) Define divergence and curl of vector field. If $\vec{V} = (y^2 - z^2 + 3yz - 2x)\,\hat{i} + (3xz + 2xy)\,\hat{j} + (3xy - 2xz + 2z)\,\hat{k}$, then find div $\vec{v}$ and cur $\vec{v}$. Determine whether $\vec{v}$ is source or sink free and rotational or irrotational. 13

8(a) Show that $\vec{V} = 2xyz\,\hat{i} + (x^2 z + 2y)\,\hat{j} + x^2 y\,\hat{k}$ is irrorational and find a scalar function U (x, y, z) such that $\vec{V} = \text{grad}\, u$. 12

8(b) Evaluate $\int_C \vec{F}.d\vec{r}$ where $\vec{F} = 3x^2\,\hat{i} + (2xz - y)\,\hat{j} + z\,\hat{k}$ along the curve, $C: x = 2t^2,\ y = t,\ z = 4t^2 - t$ from t=0 to t=1. 12

8(c) If a force $\vec{F} = 2x^2 y\,\hat{i} + 3xy\,\hat{j}$ displaces a particle in the xy-plane from (0, 0) to (1, 4) along a curve $y=4x^2$. Find the work done. 11

Department of Leather Engineering

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B. Sc. Engineering 2nd Year 1st Term Examination-2023

HUM 2119

Accounting and Industrial Law

Time: 3.0 Hours Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION–A

1(a) Abir Singha, a first accounting year student, believes that debit balances are favorable and credit balances are unfavorable. Is Abir correct? Discuss. 07

1(b) On April 1, 2024, "Patwary traders" began operations. The following transactions were completed during the month:

April 1: Owner invested BDT 400,000 in the business.

April 3: Purchased goods for cash BDT 52,500

April 5: BDT 6,000 was stolen from the cash box.

April 7: Paid salaries and wages to employes BDT 24,000.

April 9: Sold goods for cash BDT 65,000.

April 11: Purchased equipment for BDT 70,000 paying cash of BDT 61,000 and the balance was on account.

April 13: Sold goods and received a cheque of BDT 72,000.

Required:

(**i**) Show journal entries of the above transections of April 14

(**ii**) Prepare a tabular analysis of the April transections. 14

2(a) Define accounting. Describe accounting cycle with example. 15

2(b) Prepare a trial balance as on 30th June 2023 using the following Ledger balances of "LE Enterprise": 20

Goodwill 15,000; Drawings 15,000; cash in hand (01-07-2022) 50,000; Sales 1,44,000; Returns outward 500; Inventory (30-06-2023) 15,000; Accounts payable 7,000; Purchase 74,000; Allowance for bad debt 1,200; Carriage Inward 1,250; carriage outwards 2,000; Salaries Expense 5,000; Cash in hand (30-06-2023) 52,500; Rent expense 1,500, Differed Advertisement 13,000; Insurance premium 8,750; Capital 38,000; Inventory (01-07-2022) 12,000; Bank overdraft 9,300.

3(a) Describe the importance of cost accounting for manufacturing organizations. 11

3(b) Felde Bucket Co., a manufacturer rain barrels, had the following data for 2023: 24

Sales	2500 units
Sales Price	\$ 40 per unit
Variable Costs	\$ 24 per unit
Fixed Cost	\$ 19,500

Required:

(i) Compute the contribution margin ratio.

(ii) Compute the break-even point in units and in dollars.

(iii) Compute the margin of safety in dollars and as a ratio.

(iv) If the company wishes to increase its total dollar contribution margin by 40% in 2024, by how much will it need to increase its sales if all other factors remain constant?

4(a) SAFE 'N' SAVE, a super market, started operations in Khulna in 2004. The following trial balance and adjustment were taken from its accounting records of 2023. 35

SAVE 'N' SAFE

Trial Balance

As on 31 December,2023

Account Titles	Debit Tk.	Credit Tk.
Cash	13,350	
Inventory	30,760	
Office Supplies	755	
Prepaid Insurance	935	
Store Equipment	22,410	
Accumulated Depreciation-Store Equipment		3,120
Office Equipment	5,210	
Accumulated Depreciation-Office Equipment		1,130
Accounts Payable		895
Capital and Drawings	15,000	60,585
Sales		1,81,240
Sales and Purchase Returns	510	1,830
Purchases	1,04,000	
Carriage Inward	1,670	
Salaries	31,950	
Rent Expenses	9,000	
Utilities	4,600	
Wages	8,650	
	2,48,800	2,48,800

Required: Prepare:-
(i) Statement of Comprehensive Income
(ii) Statement of owner's Equity.
(iii) Statement of Financial Position.

Adjustments:

(i) Ending Inventory Tk. 25,000
(ii) Office Supplies Inventory Tk. 185 on hand
(iii) Expired Insurance Tk. 780
(iv) Utilities Expenses due Tk. 640
(v) Estimated depreciation on store equipment Tk.1,515 and on office equipment Tk. 510.

SECTION–B

5(a) Prescribe the provisions of punishment for conviction and misconduct according to Bangladesh Labor Act (BLA)-2006. 15

5(b) Explain 'Certificate of fitness' as prescribe in BLA-2006. 10

5(c) Define according to BLA 2006: **(i)** Commercial establishment **(ii)** Shift **(iii)** Lay-off and **(iv)** Labor court. 10

6(a) State the objectives of industrial law. 10

6(b) Explain the conditions of casual leave, sick leave and weekly holidays for an industrial establishment. 10

6(c) Define labor appellate tribunal. Discuss the registration cancellation process of a trade union. 15

7(a) Briefly mention different forms of unfair labor practices on the part of employers. 10

7(b) Describe working hours and leaves for workers according to shop and Establishment Act. 10

7(c) Explain the functioning and formation of labor courts. 15

8(a) Define ILO. Explain the functions of ILO in Bangladesh. 10

8(b) Write notes on the working conditions in relation to health and hygiene in RMG sector of Bangladesh. 10

8(c) What provisions should be followed to comply 'First aid appliances' and 'Canteens'? Describe. 15

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B. Sc. Engineering 2nd Year 1st Term Examination-2023

Ch 2119

Physical Chemistry

Time: 3.0 Hours — Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a) State Raoult's law for lowering of vapor pressure. 12

1(b) Why the vapor pressure of a solution of non-volatile solid is lower than that of the solvent? Explain it from the kinetic theory. 09

1(c) Discuss some practical applications of boiling point elevation and freezing point depression of solutions. 08

1(d) When 1.2 g of an organic compound (A) was dissolved in 51.0 g benzene, the freezing point depression was found to be 0.001°C. Calculate the molar mass of the compound (A). [Kf=5.12°C/mol]. 06

2(a) What are fluorescence and phosphorescence? Discuss these two processes with the help of Jablonski diagram. 10

2(b) State and explain Einstein's law of photochemical equivalence. 08

2(c) Illustrate the following terms with examples: **(i)** Quantum Yield **(ii)** Photosensitized reaction **(iii)** Quenching of fluorescence 12

2(d) Calculate energy associated with **(i)** One photon **(ii)** One Einstein of radiation of wavelength 6000A°, $h = 6.62 \times 10^{-27}$ erg-sec, $c = 3 \times 10^{10}$ $cmsec^{-1}$ 05

3(a) What is meant by adsorption? Distinguish between physical adsorption and chemisorption. 08

3(b) Deduce Langmuir adsorption isotherm and discuss this equation for the limiting condition of very low and very high pressure. 12

3(c) Write down the use of ion-exchange adsorption process for water softening and deionization of water. 10

3(d) Why does adsorption occur at the surface? 05

4(a) What are the differences between lyophilic and lyophobic colloids? 08

4(b) Write down the application of colloids in artificial kidney machine. 09

4(c) What is zeta potential? Micelle behaves like both colloid and electrolyte - explain briefly. 10

4(d) Describe the preparation of gold sol and ferric hydroxide sol. 08

SECTION–B

5(a) Write down the Debye-Huckel conductance equation for electrolyte solution and draw a schematic plot of Λ vs $\sqrt{c}$ for kcl, CH_3COOH and Na_2SO_4 solution. 10

5(b) State the Kohlrausch law of independent migration of ions. Explain the exceptionally high ionic mobility of H^+ (aq) and OH^- (aq) ions.in solution. 09

5(c) Briefly explain the 'Relaxation effect' and 'Electrophoretic effect' that winder the movements of ions. 08

Department of Leather Engineering

Khulna University of Engineering & Technology

B. Sc. Engineering 2nd Year 1st term Examination-2023

ME 2119

Basic Mechanical Engineering

Time: 3.0 Hours Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION–A

1(a) What are the two approaches of thermodynamics? Explain. 06

1(b) What is PMMI? Why it is impossible? 05

1(c) Show that any quantity of heat given to a thermodynamic system will be used to increase its internal energy and the work done by the system. 09

1(d) A system contains $0.25m^3$ of a gas at a pressure of 3.9 bar and 145°C. It is expanded adiabatically till the pressure falls to 1 bar. The gas is then heated at a constant pressure till its enthalpy increases by 72kJ. Determine the work done consider Cp = 1kJ/kgK and Cv = 0.714kJ/kgK. 15

2(a) State the Kelvin-Planck and Clausius statement of the second law of thermodynamics. Explain the equivalence of these statements. 10

2(b) From the steady flow energy equation; derive the expression for exit velocity from a nozzle in terms of enthalpy, inlet and outlet area and specific volume of the fluid. 09

2(c) Air flow steadily at the rate of 0.45kg/sec through a compressor entering at 6 m/sec with a pressure of 1 bar and a specific volume of 0.85 m^3/kg and leaving at 4.5 m/sec with a pressure of 7.0 bar and specific volume of 0.16 m^3/kg. The internal energy of the air leaving is 89 kJ/kg greater than that of air entering. Cooling water in a jacket surrounding the cylinder absorbs heat from the air of the rate of 58watt. Calculate the power required to derive the compressor and the inlet to outlet cross-sectional areas. 16

3(a) Define source and sink in the light of thermodynamics. Why a ship is unable to run by taking energy from the sea water? 07

3(b) State and prove Clausius inequality principle. 12

3(c) Define refrigerator and heat pump from thermodynamic view point. "Using a heat pump is more advantageous than direct heating". Justify the statement. 08

3(d) Define entropy from corollary 7 and show that the entropy of the universe is increasing toward a maximum. 08

4(a) What is air standard cycle? Define the following terms: **(i)** Stroke length, **(ii)** Clearance volume, **(iii)** Swept volume, **(iv)** Compression ratio, and **(v)** MEP 10

4(b) Derive the expression for efficiency of Dual combustion cycle in terms of varies ratios. 10

4(c) An engine operating on air standard Diesel cycle the compression ratio is 16 and at the beginning of isentropic compression the temperature is 23°C and pressure is 90 kPa. Heat is added until the temperature at the end of constant pressure is 1550°C. Calculate- (**i**) Cut off ratio, (**ii**) Cycle efficiency, and (**iii**) mep. 15

SECTION–B

5(a) Draw the temperature vs total heat graph during steam formation and mention the different phases of fluid. 10

5(b) What are the different features of water tube boiler and fire tube boiler? 07

5(c) Describe with neat sketches the working principle of a vertical multitubular boiler. 12

5(d) Define the terms: Dryness fraction, Enthalpy of steam, Efficiency of boiler. 06

6(a) Differentiate between boiler mountings and boiler accessories. 06

6(b) Write down the functions of the followings: (**i**) Economizer, (**ii**) Fusible plug, and (**iii**) Safety valve 06

6(c) With neat sketch describe the working principle of steam stop valve. 06

6(d) A steam plant having a boiler and superheater has the following particulars. 17
Steam pressure 13 bar, mass of stems generated = 4300 kg/h, mass of coal used= 650 kg/h, calorific value of coal= 29000 kJ/kg of coal, temperature of the feed water entering the boiler = 40°C, Dryness fraction of steam leaving the ~~superheater~~ boiler = ~~310°C~~, 0.95,
Temperature of steam leaving the superheater = 25
Determine: (**i**) Overall efficiency of the plant and (**ii**) Percentage of heat utilized in boiler and superheater.

7(a) Explain the working principle of a four-stroke cycle petrol engine with necessary sketch. 12

7(b) What are the differences between petrol engine and diesel engine? 08

7(c) What are the differences between the ideal and actual indicator diagram of a diesel engine. 07

7(d) What are the functions of (**i**) Cam shaft, (**ii**) Piston rings, (**iii**) Spark plug, and (**iv**) Fuel injector? 08

8(a) What is detonation in IC engine? 05

8(b) Briefly explain the different types of scavenging in IC engine. 08

8(c) Distinguish between impulse turbine and reaction turbine in 4 aspects. 08

8(d) Explain with neat sketch the working principle of the steam turbine suitable for high head and low discharge. 14

Department of Leather Engineering

Khulna University of Engineering & Technology

B. Sc. Engineering 2nd Year 1st Term Examination-2023

LE 2101

Leather Manufacturing Technology II

Time: 3.0 Hours | Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION–A

1(a)	Define unhairing and liming. Briefly discuss the chemistry of enzymatic unhairing with its disadvantages.	12
1(b)	Concisely deliberate the thermal unhairing method.	08
1(c)	Briefly state the chemistry of oxidative method of unhairing with its advantages.	08
1(d)	Assume that you are provided a wet salted goatskin; make a recipe for your goatskin for conducting pre-soaking, soaking, and sweating method of unhairing.	07
2(a)	List out the beamhouse operations in a tabular with its optimum pH.	05
2(b)	Explain with sketches how does pH effect of swelling on collagen? Explain.	09
2(c)	Discuss with neat sketches the three-pit liming system.	10
2(d)	Define sharpening agent. List out some common sharpening agents.	04
2(e)	The isoelectric point (IEP) of collagen shifts during liming- Explain.	07
3(a)	Splitting up of collagen fibre bundle is essential in liming operation- Explain.	10
3(b)	Define reliming. Deliming is essential before chrome tanning- Explain.	05
3(c)	A wet salted goatskin weight 2.75 kg. After liming operation pelt weight was increased to 0.5% (by wt.). Calculate the amount of ammonium sulphate to be needed for deliming to remove 50% lime. Assume that after lime wash, pelt contained 1.05% lime.	08
3(d)	Briefly state the controlling parameters of liming operation.	06
3(e)	Write short notes on: **(i)** Immunization of keratin and **(ii)** Fleshing	06
4(a)	Write down the composition of dermatan sulphate proteoglycan. List out the effect of dermatan sulphate on the final leather.	09
4(b)	Define tannery waste. Discuss the effect of limed fleshing on the environment.	09
4(c)	Discuss the behaviour of collagen in acidic and alkaline solution.	06
4(d)	Define deliming agent. Explain why ammonium salts are used as deliming agent in Bangladesh.	06
4(e)	Briefly state wastes generated in liming operation.	05

SECTION–B

5(a)	Define bating. Discuss the working mechanism of enzyme during bating.	12
5(b)	Discuss the traditional source of bating agents.	07
5(c)	Write short notes on possible defects in bating.	05
5(d)	Suppose you have a piece of delimed goat skin pelt and you have to produce belt leather from it. Now design the suitable bating operation considering required factors.	06
5(e)	Acid bate has advantages over alkaline bate. - Explain.	05

6(a)	How does enzyme work in degreasing process?	05
6(b)	Explain degreasing via solvent extraction method. What are the difficulties that arise with conventional solvent extraction degreasing process? How can it be overcome?	12
6(c)	State the working mechanism of lipase in degreasing operation.	08
6(d)	Chromium salt can exhibit as cationic, anionic and nonionic state-Justify the statement.	10
7(a)	Pickling process is essential before chrome tanning. Explain.	06
7(b)	Write down the nature of swelling of the pelt in both acidic and alkaline condition with the assistance of swelling curve. Also explain which type of swelling dominates in pickling.	11
7(c)	Briefly discuss the action of acid in polypeptide chain during pickling operation.	12
7(d)	Mention the role of non-swelling acids in pickling.	06
8(a)	After pickling, sodium thiosulphate is used. Explain.	04
8(b)	Explain the reaction mechanism of basic chromium sulphate with collagen including necessary illustration.	10
8(c)	Write note on self-basifying agent. What are the roles of masking in chrome tanning?	06
8(d)	Write down the typical preparation method of basic chromium sulphate.	10
8(e)	Why basification is done after chrome tanning. Explain.	05